

Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident

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Summary

Data Scientist with a strong background in **AI Research**, with experience across **healthcare**, **NLP**, and **geoscience** domains. Builds **end-to-end ML pipelines**, conducts **statistical and spatial analysis**, and deploys models that hold up in production. Develops **agentic AI pipelines** using **RAG**, **LLMs**, and **MCP**. Published across **IEEE**, **SAGE**, and **Springer Nature**. Comfortable working across the full stack from raw data to results, and collaborating directly with domain experts. Proficient in **Python**, **SQL**, **R**, **PyTorch**, and **TensorFlow**.

Work Experience

Data Analyst

Oct 2025 - Feb 2026, Jul 2026 - Present

The City College of New York

New York, NY

- Analyze **institutional data** using **Excel**, **Tableau**, and **Python**, surfacing patterns that shape **advising and student engagement programs**.
- Conduct **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach.
- Build **analytical reports** and **dashboards** that surface trends and anomalies across departments, replacing gut-feel decisions with **data-backed ones**.

Research Assistant

Dec 2025 - Present

CUNY Research Foundation

New York, NY

- Develop an **agentic AI pipeline** using **RAG**, **LLMs**, and **MCP-connected tools** for natural language querying over complex geoscience databases.
- Bridge **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specs and iterating on pipeline outputs for interpretability and scientific relevance.
- Design the **end-to-end technical stack: vector database architecture**, domain-specific **embedding models**, **LLM optimization**, and **cloud deployment** for a production-ready geoscience research tool.

Data Evangelist Intern

Feb 2026 - May 2026

DataCamp

New York, NY

- Designed and maintained **scalable data pipelines** to collect, integrate, and validate **multi-source data** through automated scripts, APIs, and web scraping.
- Built **data applications** and **analytical workflows** powered by **LLMs**, integrating **OpenAI** and **Hugging Face** models into user-facing tools for exploring key metrics.
- Analyzed data to surface trends and key drivers, packaging findings into **visualizations** and structured reports.

Technical Skills

Languages: Python, SQL, R

Data Analysis: NumPy, Pandas, SciPy, Excel

Data Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK, Word2Vec

LLMs & Agents: LLMs, RAG, MCP, Hugging Face, ChromaDB

Database & Big Data: MySQL, MongoDB, Snowflake

MLOps & Cloud: AWS, MLflow, Git/GitHub, Docker, Vercel

Others: Data Preprocessing, Feature Engineering, Model Evaluation, A/B Testing

Education

The City College of New York

New York, NY

M.S. in Data Science and Engineering

Dec 2027

East West University

Dhaka, Bangladesh

B.S. in Computer Science and Engineering

Dec 2023

Research Experience

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- Achieved 97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

Papers Under Review

Multimodal Learning in Medical Imaging: Methods, Benchmarks, Evaluation, & Clinical Translation – *Elsevier*

- Conducted a structured review of **multimodal learning in medical imaging**, proposing a unified taxonomy while identifying key gaps in **robustness, reproducibility, and benchmark standardization** across healthcare AI systems.

Data Projects

NYC HIV/AIDS Trend Analysis & Predictive Modeling | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered **multi-year NYC Open Data** on HIV/AIDS diagnoses into analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis** across NYC, uncovering persistent geographic and demographic disparities.
- Applied **Random Forest models ($R^2 = 0.65$)** to identify high-impact predictors for public health targeting.

Global Population Growth Modeling & Forecasting | *R, Tidyverse, ggplot2, leaps, caret*

- Engineered a **longitudinal time-series dataset** from World Bank indicators across **200+ countries** and **60+ years** of demographic trends.
- Conducted multi-decade demographic analysis by region and income group, identifying divergent growth patterns.
- Developed forecasting models achieving **$R^2 = 0.755$** using **rolling-window cross-validation**.

AI/ML Projects

Deep Learning Based Wound Classification | *Python, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib*

- Improved validation accuracy from **89.8% to 98.5%** by redesigning the CNN architecture with added convolutional layers, batch normalization, dropout, and L2 regularization.
- Applied **oversampling** to correct significant class imbalance across 10 wound categories, improving fairness and reliability.
- Reduced loss to **0.1061** using EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks to control training.

Intrusion Detection System | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- Processed and analyzed 2.5+ million network traffic records** through EDA.
- Reduced memory usage by **47.5%** through numerical datatype downcasting.
- Achieved 97% median accuracy** detecting network anomalies using machine learning models.

Clinical Risk Modeling & Unsupervised Structure Analysis | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- Curated and standardized **clinical tabular data**, addressing missingness, outliers, and feature scaling.
- Developed a **logistic regression classifier** achieving **80% test accuracy** for risk prediction.
- Evaluated **K-Means clustering with PCA and t-SNE** projections to assess latent structure.